

# Incident Management Lifecycle Automation in ServiceNow

## Phase 3 – Project Design Phase

### 1. Design Objective

The design defines how requirements are represented in ServiceNow. Incident Management is the central process and related records provide the integrated service-management structure.

### 2. High-Level Architecture

| Component                  | Role                                                |
|----------------------------|-----------------------------------------------------|
| Incident Management        | Central record and lifecycle controller.            |
| Service / Service Offering | Provides affected-service context.                  |
| Assignment / Escalation    | Routes work to responsible teams and agents.        |
| Child Incidents            | Supports decomposition of complex work.             |
| Change Management          | Controls technical changes required for resolution. |
| Knowledge Management       | Provides reusable troubleshooting content.          |
| Task SLA                   | Tracks service-level commitments.                   |
| Activity History           | Provides traceability.                              |

### 3. End-to-End Process Flow

Incident Creation → Classification → Impact/Urgency Assessment → Priority → Assignment → Investigation → Escalation if required → Knowledge/Change Support → Child Incident Management → Resolution → SLA Verification → Closure.

### 4. Data Relationship Design

- Incident is associated with a service and service offering.
- Parent incident can have child incidents.
- Change request can be linked to the incident.
- Knowledge articles can be attached or created from resolution work.
- Task SLA records track time-based commitments.
- Activity history records key changes and comments.

### 5. User Interface Design

Service Operations Workspace provides access to incident details, activity, assignment and related records. Related-record sections expose child incidents, change requests, knowledge and SLA information.

## 6. Automation Design Principles

- Controlled and traceable lifecycle transitions.
- Routing based on service and ownership.
- Escalation for issues requiring additional support.
- Parent-child synchronization where configured.
- Incident-Change traceability.
- Reusable resolution knowledge.
- SLA monitoring throughout the lifecycle.

## 7. Incident Record Design

| Design Area       | Example                                             |
|-------------------|-----------------------------------------------------|
| Short Description | Unable to connect to Corporate VPN from home office |
| Service           | Remote Access                                       |
| Service Offering  | Corporate VPN                                       |
| Assignment Group  | Network                                             |
| Assigned To       | David Loo                                           |
| Parent Incident   | INC0010001                                          |
| Change Request    | CHG0030001                                          |
| Knowledge         | VPN troubleshooting knowledge                       |

## 8. State Design

| State       | Purpose                                            |
|-------------|----------------------------------------------------|
| New         | Incident has been created and requires processing. |
| In Progress | Support team is actively investigating.            |
| On Hold     | Work is temporarily paused.                        |
| Resolved    | Solution/workaround provided and documented.       |
| Closed      | Incident fully completed according to process.     |

## 9. Exception and Escalation Design

When Service Desk cannot resolve an issue, it is routed or escalated to an appropriate Level 2 team. Assignment and activity history provide evidence of the hand-off. If a technical change is required, a change request is linked.

## 10. Validation Design

- Verify parent and child incident states.
- Verify parent-resolution activity on the child incident.
- Verify change request linkage.
- Verify created and attached knowledge.
- Verify Task SLA visibility and status.
- Verify assignment group and assigned agent.
- Verify end-to-end lifecycle.